Market Risk & Pricing Engine

Technical Documentation & Methodology Reference

Version 1.0 · Python Implementation

Compiled June 03, 2026

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1. Overview & Architecture

The **Market Risk & Pricing Engine** is a Python library designed to replicate the core analytical capabilities of professional quantitative finance platforms such as Numerix, Murex, and Bloomberg MARS. It provides a unified framework for pricing and risk-managing a wide range of financial instruments — from vanilla equity options to exotic path-dependent derivatives and credit products.

Key Capabilities

Pricing Models	BSM, Black-76, Garman-Kohlhagen, Bachelier, Heston, SABR, Binomial CRR/LR,
Vanilla Options	European, American, Bermudan — 7 model choices per instrument
Barrier Options	Single knock-in/out, double barrier, partial/window barrier
Path-Dependent	Asian (arithmetic & geometric), Lookback (fixed & floating), Cliquet, Shout
Digital Options	Cash/asset-or-nothing, One-touch, No-touch, Double no-touch, Supershare
Exotic Options	Chooser, Compound, Forward-start, Power, Reset, Range Accrual
Multi-Asset	Exchange (Margrabe), Spread, Basket, Rainbow, Quanto, Himalaya, Altiplano
Variance Prods.	Variance swap, Vol swap, Gamma swap, Corridor/Conditional var swap
Fixed Income	Bonds, FRN, IRS, OIS, Basis swap, Cap/Floor/Collar, Swaption, CMS
Credit	CDS, Binary CDS, CDO tranche (LHP/Gaussian copula), CVA/DVA
FX Instruments	FX Forward, FX Option, FX Barrier, Risk reversal, Strangle, Straddle
Greeks	Full set: Delta, Gamma, Vega, Theta, Rho, Vanna, Volga, Charm, Speed, Color, Z
Risk Metrics	Historical / Parametric / MC / EVT VaR, CVaR, Component VaR, Kupiec & Christo
Stress Testing	14 historical scenarios, reverse stress, PnL attribution, Greeks ladder

Module Architecture

Module	File	Contents
models/	`black_scholes.py`	`BSM · Black-76 · GK · Bachelier · full Greeks`
	`trees.py`	`CRR · Leisen-Reimer · Trinomial trees`
	`monte_carlo.py`	`GBM paths · Heston paths · MC pricer · LSM`
	`heston.py`	`Heston semi-analytical · SABR · calibration`
	`implied_vol.py`	`IV solvers · SVI · volatility surface`
instruments/	`vanilla.py`	`European · American · Bermudan`
	`barrier.py`	`Single/double barrier · MC barrier`

	asian.py	Geometric exact · Arithmetic MC+CV
	digital.py	Cash/asset-or-nothing · Touch options
	lookback.py	Fixed/floating lookback · MC
	exotic.py	Chooser · Compound · Cliquet · Shout · etc.
	multi_asset.py	Margrabe · Kirk · Basket · Rainbow · Quanto
	variance_swaps.py	Var swap · Vol swap · Corridor var
	fixed_income.py	Bonds · IRS · Cap/Floor · Swaption
	credit.py	CDS · CDO-LHP · CVA/DVA
	fx.py	FX Forward · GK Option · Barrier · RR/STR
risk/	var.py	VaR · CVaR · EVT · Portfolio VaR · Backtesting
	stress.py	Stress scenarios · Greeks ladder · PnL explain

2. Mathematical Models

2.1 Black-Scholes-Merton (BSM)

The Black-Scholes-Merton model is the foundational closed-form solution for pricing European options on a stock paying a continuous dividend yield *q*. It assumes log-normal price dynamics with constant volatility.

Dynamics

```
dS = (r - q) S dt + sigma S dW
```

Option Price

```
C = S e^(-qT) N(d1) - K e^(-rT) N(d2)
P = K e^(-rT) N(-d2) - S e^(-qT) N(-d1)
d1 = [ln(S/K) + (r - q + sigma^2/2) T] / (sigma sqrt(T))
d2 = d1 - sigma sqrt(T)
```

Greeks (First Order)

Greek	Call	Put
Delta (dP/dS)	`e^(-qT) N(d1)`	`e^(-qT) [N(d1) - 1]`
Gamma (d²P/dS²)	`e^(-qT) n(d1) / (S sigma sqrt(T))`	`Same as Call`
Vega (dP/dsigma)	`S e^(-qT) n(d1) sqrt(T) / 100`	`Same as Call`
Theta (dP/dt) /day	`-(S e^(-qT) n(d1) sigma)/(2sqrt(T)) - r K e^(-rT) N(d2) + q S e^(-qT) N(d1)) /365`	`+r K e^(-rT) N(-d2) - q S e^(-qT) N(-d1)`
Rho (dP/dr) /1%	`K T e^(-rT) N(d2) / 100`	`-K T e^(-rT) N(-d2) / 100`

Higher-Order Greeks

Greek	Formula	Interpretation
Vanna	`-e^(-qT) n(d1) d2 / sigma`	dDelta/dVol — vol-spot sensitivity
Volga	`S e^(-qT) n(d1) sqrt(T) d1 d2 / sigma / 100`	d²Price/dVol² (Vomma) — vol convexity
Charm	`-e^(-qT) [n(d1)(2(r-q)T - d2 sigma sqrt(T)) / (2T sigma sqrt(T)) + q N(+/-d1)] /365`	dDelta/dT — delta bleed
Speed	`-Gamma / S * (d1/(sigma sqrt(T)) + 1)`	dGamma/dS — gamma of gamma
Color	`-e^(-qT) n(d1) / (2S T sigma sqrt(T)) * (1 + d1 d2) / 365`	dGamma/dT — gamma decay
Zomma	`Gamma * (d1 d2 - 1) / sigma`	dGamma/dVol
Ultima	`-Vega/sigma * (d1 d2 (1-d1 d2) + d1² + d2²)`	d³Price/dVol³

2.2 Black-76 (Futures & Forwards)

Black-76 extends BSM to options on futures/forwards. It is the market standard for pricing interest rate caps, floors, and European swaptions. The forward price *F* replaces the spot, eliminating the dividend yield.

```
C = e^(-rT) [F N(d1) - K N(d2)]
d1 = [ln(F/K) + sigma^2 T/2] / (sigma sqrt(T)); d2 = d1 - sigma sqrt(T)
Delta = e^(-rT) N(d1) [for call on futures]
```

Note: For caps/floors, each caplet uses the forward LIBOR rate as F, discounted by the zero-coupon price.

2.3 Garman-Kohlhagen (FX Options)

The Garman-Kohlhagen (1983) model adapts BSM to currency options. The foreign interest rate *r_f* plays the role of the dividend yield. Premium can be quoted in four conventions: domestic pips, % of domestic notional, % of foreign notional, or premium-adjusted delta.

```
C = S e^(-r_f T) N(d1) - K e^(-r_d T) N(d2)
d1 = [ln(S/K) + (r_d - r_f + sigma^2/2) T] / (sigma sqrt(T))
```

Delta Conventions

Convention	Formula
Spot delta	`phi * e^(-r_f T) * N(phi * d1)`
Forward delta	`phi * N(phi * d1)`
Premium-adj. delta	`Delta_spot - Premium / S`

2.4 Bachelier (Normal Model)

The Bachelier (normal) model assumes arithmetic Brownian motion rather than geometric. This makes it suitable for near-zero or negative rates (e.g., EUR rates in 2016-2022), and for pricing options on spreads.

```
dF = sigma_n dW (absolute, not relative volatility)
C = e^(-rT) [(F-K) N(d) + sigma_n sqrt(T) n(d)]
d = (F - K) / (sigma_n sqrt(T))
```

2.5 Binomial Trees

Lattice methods discretise the price process into up/down moves and price derivatives by backward induction. They naturally handle early exercise (American) and discrete barriers.

Cox-Ross-Rubinstein (CRR)

```
u = e^(sigma sqrt(dt)); d = 1/u; p = (e^((r-q)dt) - d) / (u - d)
```

Leisen-Reimer (LR)

LR improves convergence by applying the Peizer-Pratt inversion to match the risk-neutral probabilities to the normal CDF. For a given N, accuracy is $O(1/N^2)$ vs $O(1/N)$ for CRR.

```
p = h(d2, N); u = e^((r-q)dt) * h(d1,N) / h(d2,N)
```

Trinomial Tree

The trinomial tree has three branches per node (up, middle, down) and offers better convergence for barrier options where the barrier falls between nodes.

```
dx = sigma sqrt(3 dt); pu, pm, pd = quadratic functions of (r,q,sigma,dt)
```

2.6 Monte Carlo Engine

Monte Carlo simulation generates thousands of price paths to estimate expected discounted payoffs. The implementation includes variance reduction techniques for efficiency.

GBM Path Generation

```
S(t+dt) = S(t) exp[(r - q - sigma^2/2) dt + sigma sqrt(dt) Z]
Z ~ N(0,1), antithetic variates: Z and -Z paired
```

Variance Reduction Methods

Method	Description	Typical Speedup
Antithetic Variates	Use Z and -Z pairs — halves sampling variance	~35-50%
Moment Matching	Rescale Z so that sample mean=0, variance=1 exactly	~20-30%
Control Variate	Use geometric average (known analytically) as control for Asian	~60-80%
Quasi-MC (Sobol)	Use low-discrepancy sequences instead of pseudo-random	~70-90%

Longstaff-Schwartz LSM (American / Bermudan)

LSM (2001) estimates the continuation value at each exercise date by regressing discounted future payoffs on basis functions of the current stock price, allowing optimal early exercise decisions.

```
V(S_i, t) = E[disc * V(S_{i+1}, t+dt) | S_i] ~ beta_0 + beta_1 S + beta_2
S^2 + ...
```

At each step moving backwards, the holder exercises if the intrinsic value exceeds the estimated continuation value. Basis functions are Laguerre polynomials or simple monomials (degree 3 by default).

2.7 Heston Stochastic Volatility Model

Heston (1993) is the industry-standard stochastic volatility model. It captures the volatility smile by allowing variance to follow a mean-reverting CIR process correlated with the spot.

```
dS = (r-q) S dt + sqrt(v) S dW_S
dv = kappa (theta - v) dt + xi sqrt(v) dW_v
dW_S dW_v = rho dt
```

Parameters

Parameter	Symbol	Meaning

Initial variance	`v0`	Starting level of instantaneous variance
Mean-reversion speed	`kappa`	Rate at which v reverts to theta
Long-run variance	`theta`	Equilibrium level of variance (theta = sigma_LR^2)
Vol of vol	`xi`	Volatility of the variance process
Correlation	`rho`	Spot-vol correlation (typically negative: -0.3 to -0.8)
Feller condition		2*kappa*theta > xi^2 (ensures v > 0)

Semi-Analytical Pricing (Gil-Pelaez)

```
C = S e^(-qT) P1 - K e^(-rT) P2
P_j = 1/2 + (1/pi) int_0^inf Re[e^(-i phi ln K) phi_j(phi) / (i phi)] dphi
phi(u) = exp[i u (ln S + (r-q)T) + (kappa theta / xi^2)(D-alpha)T - (v0/xi^2)G]
```

Numerically integrated via scipy.integrate.quad with 500 integration points.

2.8 SABR Model

SABR (Hagan et al. 2002) is the market-standard model for interest rate volatility surfaces. It provides a closed-form approximation for implied volatility as a function of strike, making calibration highly efficient.

```
dF = alpha F^beta dW_F
dalpha = nu alpha dW_alpha, dW_F dW_alpha = rho dt
```

Implied Volatility Approximation (ATM)

```
sigma_ATM ~ alpha / F^(1-beta) * [1 + ((1-beta)^2/24 * alpha^2/F^(2-2beta) + rho*beta*nu*alpha/(4 F^(1-beta)) + (2-3rho^2) nu^2/24) T]
```

For away-from-the-money strikes, the full formula includes a z/chi(z) correction term.

Parameter	Role
`alpha`	Initial vol level (ATM vol proxy when beta=1)
`beta`	CEV exponent: 0=Normal, 0.5=CIR-like, 1=Log-normal
`rho`	Correlation — controls skew (negative = put skew)
`nu`	Vol of vol — controls smile curvature

2.9 Implied Volatility Solvers

Implied volatility is extracted from market prices by inverting the pricing formula. The engine uses Newton-Raphson with a Vega-based update step, falling back to Brent's method if convergence fails.

```
sigma_{n+1} = sigma_n - (C(sigma_n) - C_market) / Vega(sigma_n)
Fallback: Brent bisection on [sigma_lo, sigma_hi] with tol=1e-8
```

Initial guess uses the Brenner-Subrahmanyam approximation: sigma ~ sqrt(2 |ln(S/K) + 2rT| / T).

The **SVI (Stochastic Volatility Inspired)** parameterisation fits the full vol smile slice:

```
w(k) = a + b [rho (k-m) + sqrt((k-m)^2 + sigma^2)], k = ln(K/F), w =
sigma_imp^2 T
```

3–9. Instrument Pricing

3. Vanilla Options

All vanilla options support 7 model backends:

Model key	Method	Notes
`bsm`	Black-Scholes	Exact, with dividends (default)
`black76`	Black-76	Spot treated as forward F
`gk`	Garman-Kohlhagen	FX: q = r_f
`bachelier`	Normal model	For near-zero rates
`binomial`	CRR tree N=500	Handles American & Bermudan
`binomial_lr`	LR tree N=501	Higher accuracy than CRR
`trinomial`	Trinomial N=300	Better convergence for barriers
`mc`	Monte Carlo	With antithetic + moment matching
`lsm`	LSM 50k paths	American/Bermudan only

4. Barrier Options

Barrier options are path-dependent: they are activated (knock-in) or deactivated (knock-out) when the underlying reaches a barrier level H.

Reiner-Rubinstein Closed Form

```
Price = A - B + C - D + F (exact combination depends on K vs H and in/out type)
mu = (b - sigma^2/2) / sigma^2; lambda = sqrt(mu^2 + 2r/sigma^2)
x1 = ln(S/K)/(sigma sqrt(T)) + (1+mu) sigma sqrt(T)
y1 = ln(H^2/(SK))/(sigma sqrt(T)) + (1+mu) sigma sqrt(T)
```

Barrier Types

Type	Trigger	Effect
Down-and-Out	S falls to H	Option dies, pays rebate
Down-and-In	S falls to H	Option activates
Up-and-Out	S rises to H	Option dies, pays rebate
Up-and-In	S rises to H	Option activates
Double KO	S < L or S > U	Option dies (Ikeda-Kunitomo series)
Partial	Subperiod only	Barrier active only during [t1, t2]

Window	t_start to t_end	Time-windowed monitoring (MC)

Knock-in + Knock-out = Vanilla (put-call parity for barriers)

5. Asian Options

Asian options pay based on an average of the underlying price over the life of the option, reducing volatility exposure and manipulation risk at expiry.

Geometric Average — Closed Form (Kemna-Vorst)

The geometric average is log-normally distributed, enabling exact pricing via adjusted BSM:

```
sigma_G = sigma / sqrt(3) (continuous), sigma_G = sigma sqrt((n+1)(2n+1)/(6n^2)) (discrete)
b_G = (r-q-sigma^2/2)(n+1)/(2n) + sigma_G^2/2
```

Arithmetic Average — Monte Carlo with Control Variate

```
Arith. payoff = max(A_arith - K, 0); control = geom avg (known analytically)
PV_corrected = PV_arith - beta * (PV_geom_sim - PV_geom_CF)
beta = Cov(PV_arith, PV_geom) / Var(PV_geom)
```

Averaging: Fixed strike (vs K) or Floating strike (vs S_T). Both supported.

6. Digital Options

Instrument	Payoff	Method
Cash-or-Nothing	`cash * 1{S_T > K}`	`e^(-rT) N(d2) * cash`
Asset-or-Nothing	`S_T * 1{S_T > K}`	`S e^(-qT) N(d1)`
Gap Option	`(S_T - K2) * 1{S_T > K1}`	`BSM-style with two strikes`
One-Touch	`cash if H ever touched`	`Reiner-Rubinstein rebate formula`
No-Touch	`cash if H never touched`	`Bond - One-touch`
Double No-Touch	`cash if L < S < U always`	`Monte Carlo simulation`
Supershare	`S_T/K_lo * 1{K_lo < S_T < K_hi`	`Difference of asset-or-nothings`

7. Lookback Options (Goldman-Sosin-Gatto)

Lookback options pay based on the extremum of the underlying price over the option's life. They have no exercise risk — the holder always gets the best possible outcome. Exact closed-form exists for European-style.

```
Floating Call: max(S_T - S_min, 0) [pays S_T minus running minimum]
Floating Put: max(S_max - S_T, 0) [pays running maximum minus S_T]
```

```
Fixed Call: max(S_max - K, 0) [call on maximum]
Fixed Put: max(K - S_min, 0) [put on minimum]
```

Floating Lookback Call (Continuous Monitoring)

```
Price = S e^(-qT) N(a1) - M e^(-rT) N(a2) + S e^(-rT) (sigma^2/2b) *
[(S/M)^(-2b/sigma^2) N(-a3) - e^(bT) N(-a1)]
a1 = [ln(S/M) + (b + sigma^2/2) T] / (sigma sqrt(T)); M = min(S_t)
observed so far
```

8. Exotic Single-Asset Options

Option	Reference	Key Formula / Method
Simple Chooser	Rubinstein 1991	C(S,K,T) + P(S, K*e^(-(r-q)(T-Tc)), Tc)
Complex Chooser	Rubinstein 1991	Bivariate normal, critical S* solves C(S*)=P(S*)
Compound	Geske 1979	Bivariate normal on critical S*; option on option
Forward-Start	—	At T_start, K := alpha * S(T_start); adjusted BSM
Shout	—	LSM: shout when intrinsic > continuation value
Power (symmetric)	—	BSM with S^n, adjusted drift and vol
Power (asymmetric)	—	MC: payoff = S_T^n * max(S_T - K, 0)
Cliquet/Ratchet	—	MC: sum of period returns R_i, capped and floored
Reset	—	MC: K resets at T_reset if OTM
Range Accrual	—	MC: coupon * fraction of days in [L, U]

9. Multi-Asset Options

Option	Payoff / Method
Exchange (Margrabe)	max(S1_T - S2_T, 0) — Margrabe (1978) bivariate BSM
Spread (Kirk)	max(S1 - S2 - K, 0) — Kirk (1995) approximation
Basket (MC)	max(sum(w_i S_i) - K, 0) — correlated GBM Monte Carlo
Basket (moments)	Levy (1992) moment-matched log-normal approximation
Best-of (n=2)	max(S1, S2, K) — Stulz (1982) exact formula
Best-of (n>2)	max(S1,...,Sn, cash) — Monte Carlo
Worst-of	min(S1,...,Sn) — Monte Carlo
Rainbow call	max(S_best - K, 0) — MC on best performer
Quanto	Domestic payoff on foreign asset; GK with drift adj rho*sigma_S*sigma_FX
Himalaya	Sum of best performers per period (removed each period) — MC

Altiplano	Full coupon if all assets above barrier, else basket return — MC

Correlated Multi-Asset GBM

```
paths = Cholesky(Corr) @ Z, Z ~ N(0, I_{n x steps})
S_i(t+dt) = S_i(t) exp[(r - q_i - sigma_i^2/2) dt + sigma_i sqrt(dt) * paths_i]
```

10–12. Fixed Income & Credit

10. Variance & Volatility Products

Variance products allow direct trading of realised volatility, independent of direction.

Product	Payoff	Fair Strike
Variance Swap	N * (RV^2 - K_var)	Model-free via log-contract replication
Vol Swap	N * (RV - K_vol)	Brockhaus-Long approx or MC
Gamma Swap	N * integral (S_t/S_0) dV_t	= sigma^2 under GBM
Corridor Var	N * sum_in_{[L,U]} r_i^2 * ann.	MC simulation
Conditional Var	E[RV | in corridor]	MC: corridor var / fraction time in

Model-Free Variance Strike (Demeterfi et al. 1999)

```
K_var = (2/T) [ln(F/S_0) - (F/S_0 - 1)] + (2/T) int_0^F (P(K)/K^2 * (1-ln(K/F))) dK
+ (2/T) int_F^inf (C(K)/K^2 * (1-ln(K/F))) dK
```

11. Fixed Income Instruments

Bond Pricing

```
P = sum_{i=1}^{n} CF_i * D(t_i), D(t) = e^(-r(t) * t) [continuous]
```

Risk Metrics

Metric	Formula
Macaulay Duration	D_mac = sum t_i * PV(CF_i) / P
Modified Duration	D_mod = D_mac / (1 + y/freq) [periodic compounding]
Convexity	C = sum t_i^2 * PV(CF_i) / P
DV01	DV01 = P * D_mod / 10000 (price change per 1bp)
YTM	Flat yield y: P = sum CF_i e^(-y t_i) (solved via Brent)
Z-spread	Spread z added to curve: P = sum CF_i D(t_i) e^(-z t_i)
Delta Price	dP/P ~ -D_mod * dy + 0.5 * C * (dy)^2 [Taylor expansion]

Interest Rate Swap (IRS)

```
NPV = N * [PV_float - PV_fixed]
PV_fixed = K * sum dt_i * D(T_i) = K * Annuity
PV_float = D(T_0) - D(T_n) [par FRN approximation]
```



```
Fair rate K* = (D(T_0) - D(T_n)) / Annuity
DV01 = N * Annuity / 10000
```

Cap / Floor Pricing (Black-76 Caplets)

```
Caplet_i = N * tau_i * D(T_i) * Black76(F_i, K, T_{i-1}, r, sigma)
Cap = sum_{i=1}^{n} Caplet_i; Floor = sum_{i=1}^{n} Floorlet_i
Collar = Cap - Floor (costless when strike of cap and floor chosen
appropriately)
```

European Swaption (Black-76)

```
Swaption = N * Annuity * Black76(S_0, K, T_opt, r, sigma, call/put)
Annuity = sum_{i=1}^{m} dt_i * D(T_opt + i/freq)
S_0 = (D(T_opt) - D(T_opt + T_swap)) / Annuity [forward swap rate]
```

12. Credit Instruments

CDS Pricing (Constant Hazard Rate)

```
Survival probability: Q(tau > T) = e^(-lambda * T)
Hazard rate approx: lambda ~ spread / (1 - R)
Premium leg PV = N * spread * sum dt_i * D(T_i) * Q(T_i) [risky annuity]
Protection leg = N * (1-R) * lambda * integral e^(-(r+lambda)t) dt
Fair spread = Protection_PV / (N * Risky_Annuity)
Risky DV01 = N * Risky_Annuity / 10000
```

CDO Tranche (Large Homogeneous Pool / Gaussian Copula)

```
Cond. default prob: p(x) = N[(N^-1(p) - sqrt(rho) x) / sqrt(1-rho)]
E[Tranche loss | x] = E[max(L-K1,0)] - E[max(L-K2,0)] via normal approx
E[Tranche loss] = integral p(x) phi(x) dx (Gauss-Hermite quadrature)
```

CVA / DVA

```
CVA = (1-R_cpty) * integral_0^T EPE(t) * lambda_cpty * e^(-(r+lambda)t) dt
DVA = (1-R_own) * integral_0^T ENE(t) * lambda_own * e^(-(r+lambda)t) dt
```

EPE = Expected Positive Exposure. DVA is the bilateral counterpart to CVA.

13–15. FX, Risk Metrics & CLI

13. FX Instruments

FX Forward

```
F = S * e^((r_d - r_f) * T)
Swap points = F - S (quoted in pips = 4th decimal place for major pairs)
```

FX Option — Garman-Kohlhagen (q = r_f)

Four premium conventions are computed simultaneously:

Convention	Formula
Domestic pips	`GK price (per 1 unit of foreign)`
Premium %domestic	`GK price / S`
Premium %foreign	`GK price / (K e^(-r_d T))`
Spot delta	`phi * e^(-r_f T) * N(phi d1)`
Forward delta	`phi * N(phi d1)`
Premium-adjusted delta	`Delta_spot - Premium / S`

Vol Surface Conventions

```
sigma_call(25D) = ATM + Strangle + RR/2
sigma_put(25D) = ATM + Strangle - RR/2
```

RR = Risk Reversal = sigma_call - sigma_put. Strangle = 0.5(sigma_call+sigma_put) - ATM.*

14. Market Risk Metrics

14.1 Value at Risk (VaR)

Method	Formula	When to Use
Historical	Percentile of sorted historical P&L distribution	Captures fat tails; no model assumption
Parametric Normal	VaR = -(mu T + z_a sigma sqrt(T))	Fast; assumes normality
Parametric t-dist	VaR using Student-t quantile with fitted df	Better fat-tail capture
Monte Carlo	Simulate P&L; percentile of simulated dist.	For non-linear portfolios
EVT / POT	Fit GPD to tail exceedances; extrapolate	For high-confidence (99%+) VaR

```
CVaR (ES) = E[Loss | Loss > VaR] = - mu*T + sigma*sqrt(T) *
n(z_a)/(1-alpha)
Component VaR_i = w_i * (Cov_i . port_vol) * z_a * N [sum = total VaR]
```

```
Horizon scaling: VaR(h days) = VaR(1 day) * sqrt(h) [square-root-of-time rule]
```

EVT — Peaks Over Threshold (GPD)

```
Threshold u = percentile(losses, (1-p_thr) * 100)
Fit Generalised Pareto: F(y) = 1 - (1 + xi y / beta)^(-1/xi) to exceedances y = L - u
VaR = u + beta/xi * ((alpha * N / N_u)^(-xi) - 1)
CVaR = (VaR + beta - xi * u) / (1 - xi) [requires xi < 1]
```

Backtesting

Test	H0	Statistic
Kupiec POF	Exception rate = 1 - alpha	LR = -2 ln[L_const / L_hat] ~ chi2(1)
Christoffersen	Exceptions are independent	LR_ind ~ chi2(1) based on pi01, pi11
Basel Traffic Light	Model acceptance	Green: exc < expected; Red: >3*expected

14.2 Stress Testing

14 calibrated historical scenarios with spot, volatility, and rate shocks:

Scenario	Spot Shock	Vol Shock	Rate Shock
Black Monday (1987)	-22.5%	+60%	+0.2%
Gulf War (1990)	-15.0%	+30%	-0.3%
LTCM / Russia (1998)	-20.0%	+50%	-1.0%
Dot-com bust (2000-2002)	-49.0%	+35%	-2.5%
9/11 (2001)	-7.0%	+30%	-0.5%
Lehman collapse (2008)	-35.0%	+80%	-2.0%
EUR Sovereign (2010-12)	-22.0%	+40%	+3.0%
Taper Tantrum (2013)	-6.0%	+25%	+1.0%
China Devaluation (2015)	-11.0%	+35%	-0.5%
COVID crash (2020-03)	-35.0%	+80%	-1.5%
Meme squeeze (2021-01)	+50.0%	+60%	0%
Rate hike shock (2022)	-18.0%	+30%	+4.0%
Bull run (generic)	+30.0%	-25%	+0.5%
Flash crash (generic)	-10.0%	+70%	0%

14.3 P&L; Attribution

Decomposes realised P&L; into contributions from each risk factor:

```
PnL = Delta * dS + 0.5 * Gamma * dS^2 + Vega * dSigma * 100
+ Theta * dt + Rho * dr * 100 + Vanna * dS * dSigma + 0.5 * Volga *
dSigma^2
```

Unexplained residual = actual P&L; minus sum of above terms (higher-order cross effects).

15. CLI Usage Guide

All functionality is accessible from the command line via **main.py**:

```
python3 main.py [--param value ...]
```

Vanilla Options

```
python3 main.py option --spot 100 --strike 105 --expiry 0.5 --vol 0.25 --type call --model bsm
python3 main.py option --spot 100 --strike 100 --expiry 1.0 --vol 0.20 --exercise american
--model lsm
python3 main.py option --spot 100 --strike 100 --expiry 0.5 --vol 0.20 --exercise bermudan
--model binomial
```

Barrier Options

```
python3 main.py barrier --spot 100 --strike 100 --barrier 85 --expiry 0.5 --barrier_type
down-out
python3 main.py barrier --spot 100 --strike 100 --barrier 90 --expiry 0.5 --barrier_type
down-in --mc
python3 main.py barrier --spot 100 --strike 100 --lower 80 --upper 120 --expiry 1.0 --double
```

Asian Options

```
python3 main.py asian --spot 100 --strike 100 --expiry 0.5 --vol 0.20 --fixings 12
python3 main.py asian --spot 100 --strike 100 --expiry 0.5 --geometric --continuous
python3 main.py asian --averaging floating --expiry 0.5 --vol 0.25
```

Digital Options

```
python3 main.py digital --digital_type cash_or_nothing --strike 100 --cash 1.0
python3 main.py digital --digital_type one_touch --barrier 110 --direction up --payment expiry
python3 main.py digital --digital_type double_no_touch --lower 90 --upper 110
```

Lookback Options

```
python3 main.py lookback --lb_style floating --type call
python3 main.py lookback --lb_style fixed --strike 100
python3 main.py lookback --lb_style floating --mc
```

Variance Swaps

```
python3 main.py variance_swap --spot 100 --vol 0.20 --expiry 1.0 --notional 1000000
python3 main.py variance_swap --spot 100 --vol 0.20 --expiry 1.0 --lower 90 --upper 110
```

Fixed Income

```
python3 main.py bond --face 100 --coupon 0.05 --expiry 10 --rate 0.04 --freq 2
python3 main.py irs --notional 10000000 --fixed_rate 0.04 --expiry 5 --rate 0.035
python3 main.py cap_floor --notional 10000000 --strike 0.05 --expiry 5 --vol 0.20
python3 main.py swaption --notional 10000000 --strike 0.04 --t_option 1.0 --t_swap 5.0 --vol 0.20
python3 main.py cds --notional 10000000 --spread 0.01 --expiry 5 --recovery 0.40
```

FX Instruments

```
python3 main.py fx_forward --spot 1.08 --r_d 0.04 --r_f 0.02 --expiry 0.25
python3 main.py fx_option --spot 1.08 --strike 1.09 --r_d 0.04 --r_f 0.02 --vol 0.08
python3 main.py fx_barrier --spot 1.08 --strike 1.09 --barrier 1.05 --barrier_type down-out
```

Risk Metrics

```
python3 main.py var --value 5000000 --confidence 0.99 --horizon 10
python3 main.py var --value 1000000 --confidence 0.95 --returns returns.csv
python3 main.py stress --spot 100 --strike 100 --expiry 0.5 --vol 0.20
python3 main.py greeks_ladder --spot 100 --strike 100 --expiry 0.5 --vol 0.20
python3 main.py pnl_explain --spot 100 --strike 100 --expiry 0.5 --ds 3.0 --dvol 0.02 --dt 1
```

Stochastic Models

```
python3 main.py implied_vol --market_price 5.0 --spot 100 --strike 100 --expiry 0.5
python3 main.py heston --spot 100 --strike 100 --expiry 0.5 --v0 0.04 --kappa 2.0 --xi 0.3 --rho_heston -0.7
python3 main.py sabr --spot 100 --strike 105 --expiry 0.5 --alpha 0.15 --beta 0.5 --nu 0.4 --rho_sabr -0.3
```

Dependencies

Package	Version	Usage
numpy	>=1.24	Array operations, random number generation
scipy	>=1.11	Optimization, integration, statistics, distributions
reportlab	>=4.0	PDF document generation (documentation only)

```
pip3 install numpy scipy
```

References

- Black, F. & Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3), 637-654.
- Black, F. (1976). The Pricing of Commodity Contracts. *Journal of Financial Economics*, 3, 167-179.
- Garman, M.B. & Kohlhagen, S.W. (1983). Foreign Currency Option Values. *Journal of International Money and Finance*, 2, 231-237.

- Heston, S. (1993). A Closed-Form Solution for Options with Stochastic Volatility. *Review of Financial Studies*, 6(2), 327-343.
- Hagan, P. et al. (2002). Managing Smile Risk. *Wilmott Magazine*, 84-108. [SABR model]
- Cox, J., Ross, S. & Rubinstein, M. (1979). Option Pricing: A Simplified Approach. *Journal of Financial Economics*, 7, 229-263. [CRR]
- Longstaff, F. & Schwartz, E. (2001). Valuing American Options by Simulation. *Review of Financial Studies*, 14(1), 113-147. [LSM]
- Margrabe, W. (1978). The Value of an Option to Exchange One Asset for Another. *Journal of Finance*, 33(1), 177-186.
- Reiner, E. & Rubinstein, M. (1991). Breaking Down the Barriers. *Risk Magazine*, 4(8), 28-35.
- Kemna, A.G.Z. & Vorst, A.C.F. (1990). A Pricing Method for Options Based on Average Asset Values. *Journal of Banking and Finance*, 14, 113-129. [Asian]
- Demeterfi, K. et al. (1999). A Guide to Volatility and Variance Swaps. *Journal of Derivatives*, 6(3), 9-32.
- Goldman, B., Sosin, H. & Gatto, M. (1979). Path Dependent Options: Buy at the Low, Sell at the High. *Journal of Finance*, 34, 1111-1127. [Lookback]
- Kirk, E. (1995). Correlation in the Energy Markets. In V. Kaminski (Ed.), *Managing Energy Price Risk*. Risk Publications.
- Geske, R. (1979). The Valuation of Compound Options. *Journal of Financial Economics*, 7(1), 63-81.
- Kupiec, P. (1995). Techniques for Verifying the Accuracy of Risk Measurement Models. *Journal of Derivatives*, 3(2), 73-84.
- Christoffersen, P. (1998). Evaluating Interval Forecasts. *International Economic Review*, 39(4), 841-862.
- Stulz, R. (1982). Options on the Minimum or Maximum of Two Risky Assets. *Journal of Financial Economics*, 10, 161-185.